On the Proximity of Cosmic Mass–Radius Ratios to Black Hole Limits

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Abstract

This study examines how the sizes and masses of cosmic objects—such as stars and planets—relate to the extreme limits that could cause them to collapse into black holes. Black holes are characterized by a critical boundary known as the Schwarzschild radius, beyond which **gravity** becomes so strong that nothing, not even light, can escape. A natural question arises as to whether these black-hole limits can provide insight into ordinary cosmic objects that exist far from such extreme conditions.

To investigate this, we first apply simplified Newtonian physics, typically introduced at the high-school level. While this approach is mathematically straightforward, it can yield misleading results because it does not fully incorporate the effects predicted by Einstein's theory of relativity. Relativistic effects become significant when **gravity** is exceptionally strong or when matter reaches extreme densities.

We then adopt a more rigorous, calculus-based framework that accounts for gravitational effects with greater accuracy. This method produces results that align more closely with relativistic expectations for massive stars and black-hole formation.

Overall, this work highlights the interplay between classical physics and relativity in determining the physical limits of cosmic objects. It clarifies the distinction between ordinary stellar structures and the extreme conditions required for black-hole collapse, offering a clearer perspective on how **gravity** governs structure and stability across the universe.

1 Introduction

1.1 Background and Motivation

The Schwarzschild radius emerges naturally in general relativity as a critical scale associated with gravitational collapse. It defines the radius at which the escape velocity from a spherically symmetric mass equals the speed of light, marking the formation of an event horizon. This simple mass-radius relationship is often presented as a clear boundary between ordinary astrophysical objects and black holes.

$$r_s = \frac{2GM}{c^2} \tag{1}$$

where G is the gravitational constant, M is the mass of the object, and c is the speed of light. This simple mass-radius relationship is often presented as a clear boundary between ordinary astrophysical objects and black holes.

The apparent simplicity of this criterion makes it tempting to apply it broadly. In particular, expressing the Schwarzschild condition as an inequality involving mass and radius suggests that one might compare ordinary cosmic objects—such as stars or planets—against black hole limits in a direct and quantitative way. Initially, this motivates the idea that meaningful physical insight could be obtained by examining how close such objects come to satisfying this condition.

However, this intuition raises an important question: does mathematical proximity to the Schwarzschild limit necessarily imply physical relevance, or does such reasoning obscure deeper constraints imposed by gravitational theory?

1.2 Limitations of Naive Schwarzschild Arguments

The Schwarzschild radius is derived under assumptions such as spherical symmetry, vacuum exterior, and full general relativity. Ignoring these assumptions can produce arguments that are mathematically correct but physically misleading.

In Newtonian terms, the escape velocity from an object is

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}, \tag{2}$$

and setting $v_{\text{esc}} = c$ formally gives

$$R = \frac{2GM}{c^2} = r_s, \tag{3}$$

explaining the intuitive, classical motivation. However, this neglects relativistic effects, internal pressure, and spacetime curvature, making collapse physically impossible for ordinary objects.

This tension between mathematical validity and physical meaning motivates the present study, which examines the limits of applying Schwarzschild-based criteria.

1.3 Scope and Contributions of This Work

This work investigates the conditions under which Schwarzschild-based mass-radius comparisons retain physical meaning and identifies regimes where such comparisons break down. While the Schwarzschild radius provides a precise boundary for black hole formation, its application to ordinary stars and planets can be misleading when naive Newtonian intuition is applied.

For example, consider a neutron star with mass $M \sim 2M_\odot$ and radius $R \sim 12$ km. Using the Newtonian escape velocity formula,

$$v_{\rm esc} = \sqrt{\frac{2GM}{R}} \approx 0.6\,c,$$

one might conclude that this object is dangerously close to becoming a black hole. However, fully relativistic models show that the internal pressure and equation of state prevent collapse, illustrating the gap between mathematical suggestion and physical reality.

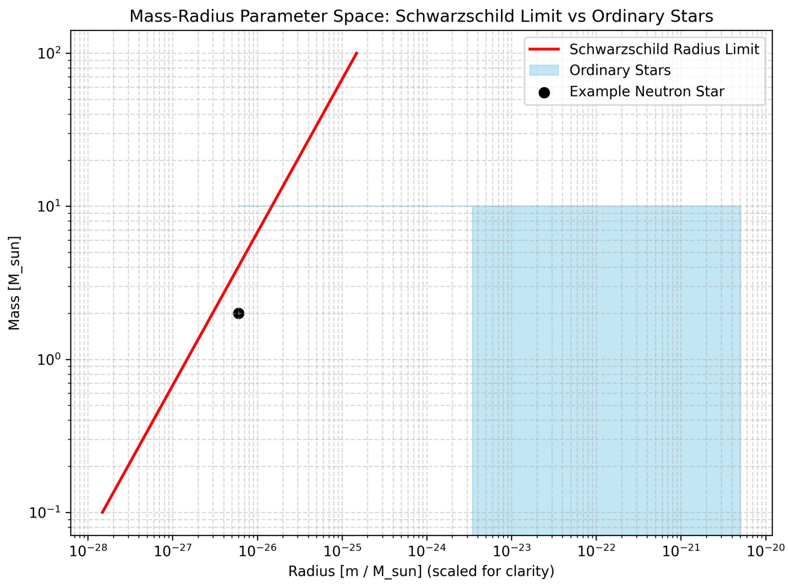

Figure 1: Schematic illustration of the mass–radius parameter space. The Schwarzschild radius boundary (red line) marks the theoretical limit for black hole formation. Ordinary stars (blue region) lie safely below this limit, highlighting the gap between Newtonian intuition and relativistic reality. Some supporting computations for this figure were generated using Python scripts, with plotting handled via Matplotlib and Plotly.

By contrasting Newtonian approximations with calculus-based and relativistically informed analyses, this study systematically explores the mass–radius parameter space of non-collapsed cosmic objects. Computational methods, including numerical evaluation of compactness and visualization of parameter regimes, are employed to distinguish physically meaningful results from misleading interpretations.

2 A Naive but Misleading Schwarzschild Reformulation

2.1 Algebraic Reformulation

The Schwarzschild radius is defined by the condition

$$\frac{2GM}{R} = c^2, \tag{4}$$

which may be rearranged as

$$2GM = c^2 R. \tag{5}$$

Introducing a formal kinematic interpretation, we write one factor of the speed of light as a ratio of differentials,

$$c = \frac{dx}{dt}, \tag{6}$$

while retaining the second factor explicitly. Substituting into the previous expression yields

$$2GM = cR\frac{dx}{dt}. \tag{7}$$

Multiplying both sides by dt, we obtain

$$2GM\,dt = cR\,dx. \tag{8}$$

2.2 Hypothetical Photon-Transit Interpretation and Black Hole Hypothesis

Equation (8) can be interpreted under a formal but physically naive hypothesis. Let x denote the total coordinate distance traversed by a photon along a straight radial path. Points A and B denote the start and end of the path, respectively, while O marks the entry point of a black hole. Inside the event horizon ($x \geq O$), the physics is unknown and potentially dominated by extreme tidal forces and spaghettification, which stretch objects along the radial direction.

For simplicity, we assume a symmetric division of the trajectory such that

$$OA = OB = \frac{x}{2}.$$

2.3 Formal Integration and Resulting Timescale

Integrating both sides of Equation (8), we write

$$\int_0^T 2GM\,dt = \int_{x/2}^{x} cR\,dx. \tag{9}$$

The choice of integration limits reflects the assumptions made in this naive construction. On the left-hand side, we integrate over t from 0 to T, corresponding to the photon being emitted at point A and reaching point B along the trajectory. On the right-hand side, we integrate over x from $x/2$ to x, reflecting the symmetric division

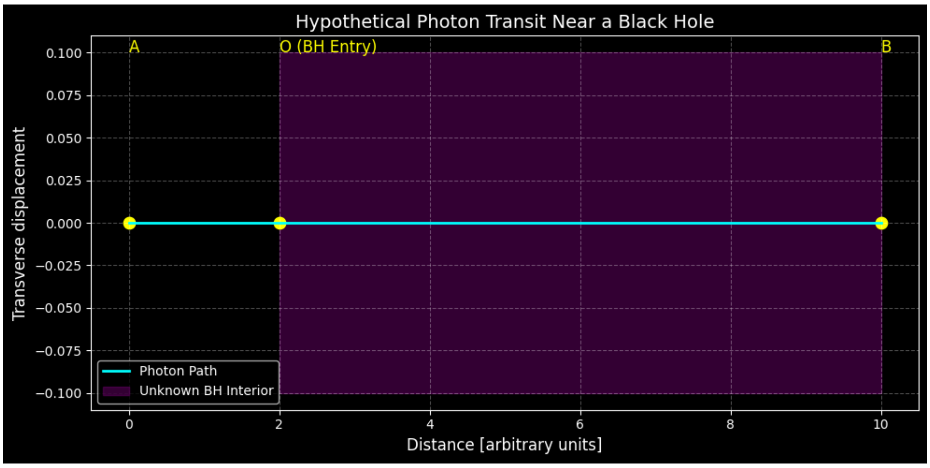

Figure 2: Hypothetical photon trajectory near a black hole. The cyan line represents the photon path. The purple shaded region marks the black hole interior where physics is unknown, potentially dominated by spaghettification. O is the entry point of the black hole. This figure was generated programmatically using Python and Matplotlib. Minor computational steps involved Plotly and Manim for related illustrations.

of the path around the entry point O of the black hole. This implicitly assumes that the photon travels along a straight radial line at constant speed c, both outside and (naively) inside the event horizon. While this choice is mathematically convenient, it does not correspond to any physically valid trajectory in general relativity, where spacetime curvature and tidal effects dominate.

Evaluating the integrals gives

$$2GM\,T = cR\left(x - \frac{x}{2}\right) = \frac{cRx}{2}, \tag{10}$$

which can be rearranged to

$$4GM\,T = cRx. \tag{11}$$

While this derivation is mathematically consistent, it is physically naive: it ignores relativistic effects, the curvature of spacetime, and the impossibility of classical photon trajectories inside the event horizon. Therefore, despite its formal correctness, the result does not describe real black hole physics.